

Logical Bilateralism - Proofs, Models and Applications

Session 2 – Classical frameworks: Signed bilateralist natural deduction and multi-conclusion sequent calculus

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Recap from Session 1

Topic: Foundations: Models, Natural Deduction, and Sequent Calculus

- Natural deduction proofs, harmony for the operational rules, and normalisation.
- Single-conclusion sequent calculus, motivating sequents as explicit bookkeeping for natural deduction proofs; relation between cut-free derivations and normal proofs
- Multiple-conclusion classical sequent calculus
- Models for classical and intuitionist logic
- Question about the asymmetry between truth and falsity or premise and conclusion in models and proofs

Today's Session

The original motivation of bilateralism and its road to classical logic.

- Harmony as a challenge for a classical proof-theoretic semantics
- Assertion and denial as primitive
- Signed formulas in a natural deduction approach
- A reading of the sequent calculus approach

Background: Proof-Theoretic Semantics

Proof-theoretic semantics in the philosophical tradition:

Denotationalism

- Meaning of linguistic expressions is determined by what the expressions denote, by their reference
 - Truth and meaning are in principle *independent* of us

Inferentialism

- Meaning of linguistic expressions is determined by how the expressions are used in inferences (Brandom 2000)
 - Drawing inferences as a fundamentally human activity
- Wittgensteinian “**Meaning is Use**” tradition

For a large class of cases of the employment of the word ‘meaning’—though not for all—this word can be explained in this way: the meaning of a word is its use in the language (Wittgenstein PI 43)

Proof-Theoretic Semantics

Proof-Theoretic Semantics

The meaning of logical terms is given by the proof rules used in logic. (Note: Term “Proof-theoretic semantics” was only coined in the 1990s but idea was prevalent much longer)

General Proof Theory

PTS belongs to the broader program of *General Proof Theory*, developed in the 1960/70s by Prawitz, Dummett, Martin-Löf, Kriegl, etc. on the basis of Gentzen’s work.

Gentzen’s remarks on his calculus of natural deduction:

The introductions represent, as it were, the ‘definitions’ of the symbols concerned, and the eliminations are no more, in the final analysis, than the consequences of these definitions. (Gentzen 1964[1935], p. 295)

The original idea of proof-theoretic semantics

Rudolf Carnap (1934) in “The Logical Syntax of Language” about the relation between meaning and inference rules:

- Usually we ask: given that $\wedge$ means ‘and’, what rules (should) govern $\wedge$ in valid inferences?
- ⇒ Presupposes some antecedent meaning on which basis we create inference rules
- Instead: we should state inference rules, i.e., which propositions allow us to *derive* sentence $A \wedge B$ and what can be *derived from* this sentence, they will determine a meaning
- ⇒ Rules yield meaning, not the other way around
- ⇒ Rules need no further justification
- ⇒ It’s part of the meaning of $\wedge$ that from $A \wedge B$ we get A (and B)

A challenge for proof-theoretic semantics

A famous objection: Prior's tonk

Formulated in Arthur Prior's (1960) "The Runabout Inference-Ticket" in the form of an *ad absurdum* argument:

If all we need to define the meaning of a connective is giving rules, then why not consider this one:

It's called tonk.

What is its meaning?

Simple! These rules:

$$\frac{A}{A \text{ tonk } B} \text{ tonk}I \qquad \frac{A \text{ tonk } B}{B} \text{ tonk}E$$

(Observe how awfully similar those look to the following rules:)

$$\frac{A}{A \vee B} \vee I \qquad \frac{A \wedge B}{B} \wedge E_2$$

A famous objection: Prior's tonk

$$\frac{A}{A \text{ tonk } B} \text{ tonk } I \qquad \frac{A \text{ tonk } B}{B} \text{ tonk } E$$

Why is tonk problematic?

Because with tonk we can derive everything from anything:

$$\frac{\frac{A}{A \text{ tonk } B} \text{ tonk } I}{B} \text{ tonk } E$$

⇒ Trivializes system

This means that not *any* kind of rules can be considered meaning-conferring

Consequences for PTS

- `tonk` shows that not all rule sets yield meaningful or valid connectives
- From this point, one core task of PTS became: **How to distinguish proper from improper rules?**
- What requirements must rules for connectives meet in order to be meaning-conferring?
- How can we keep the ‘good’ rules (conjunction etc.) but dismiss the ‘bad’ ones (the `tonk`-ish)?

The Quest for Harmony

The Principle of Harmony

Informally: Some kind of balance between introduction and elimination rules which somehow ensures an exclusion of tonk -like rules (Tennant 1978)

⇒ Rules that are in harmony are ‘safe’

What is the harmony principle formally? (Francez/Dyckhoff 2012)

- Conservativity constraint (Belnap 1962), **total harmony** (Dummett 1975)
- Normalizability constraint, **local harmony** (Dummett 1991)
- Inversion principle (Prawitz 1965, 1971), **general-elimination harmony** (Read 2010)

The Principle of Harmony

Inversion Principle

We shouldn't be able to derive more from an elimination rule whose major premise is the conclusion of an introduction rule than what was already established by that introduction rule.

$$\frac{\frac{A}{A \wedge B} \wedge I}{A} \wedge E_1$$

$$\frac{\frac{A}{A \wedge B} \wedge I}{B} \wedge E_2$$

$$\frac{\frac{A}{A \text{ tonk } B} \text{ tonk } I}{B} \text{ tonk } E$$

⇒ Reductions make inversion principle explicit

The Principle of Harmony

Classical negation rules (natural deduction):

$$\begin{array}{c} [A] \\ \vdots \\ \# \\ \hline \neg A \end{array} \neg I \qquad \frac{\neg A \quad A}{\#} \neg E \qquad + \qquad \begin{array}{c} [\neg A] \\ \vdots \\ \# \\ \hline A \end{array} \neg E \qquad \text{or} \qquad \frac{\neg \neg A}{A} \neg E$$

Intuitionistic rules

$$\begin{array}{c} [\neg A] \\ \vdots \\ \# \\ \hline \neg \neg A \end{array} \neg I \qquad \frac{\neg \neg A}{A} \neg E$$

Reductions?

Debate about intuitionistic vs. classical logic

Why is PTS leaning towards intuitionistic logic?

- Preference for natural deduction systems, which for IL satisfy certain PTS-desirable properties that CL systems do not:
 - Normalization
 - Subformula property
 - Introduction form property

⇒ PTS seems to distinguish intuitionistic logic

⇒ Classical logic looks structurally defective from a PTS point of view

Bilateralism to the rescue!

Bilateralist Conception of Meaning

- Understanding a sentence A : knowing how to answer, *yes* or *no*, to the question “Is it the case that A ?”
- Example: understanding “It is raining” requires knowing
 - the conditions for answering *yes*, **and**
 - the conditions for answering *no*.

Rumfitt (2000):

“Affirmative answers, however, enjoy no priority over negative ones. The acts of answering a propositional question affirmatively and of answering it negatively - the acts of accepting its content and of rejecting it - are conceived to be on all fours. I shall call this view of how a sentence’s sense might be specified the *bilateral* conception, in contradistinction to the *unilateral* conception”

Definitions of bilateralism

What most characterizations have in common:

- Reference to speech acts of assertion and denial
- (But other approaches also exist, taking truth/falsity or provability/refutability on a par)
- Emphasizing those two notions' symmetry ("both being primitive", "not reducible to each other", "being on a par",...)
- Another point often mentioned: In bilateralism rejection and/or denial are considered as conceptually prior to negation

⇒ Against Frege: denial of A is asserting $\neg A$

Two main directions in bilateralism: Same motivation, different implementations

The signed-formula approach

Seminal Paper: Rumfitt (2000): 'Yes' and 'No'

- Starting Point: Dummett's critique of CL with respect to PTS: rules in classical ND are not in harmony!
- Rumfitt against this:
 - Based on wrong (Fregean) conception of meaning-as-use
 - Consider speech acts of assertion *and* denial in rules, then you get a system of CL, which is harmonious
- Rumfitt encodes speech acts as **signed formulas**:
 - $+A$:= " A is asserted"
 - $-A$:= " A is denied"
- Natural deduction rules now govern signed formulas
- For each connective we need:
 - introduction / elimination rules for $+$ (assertion)
 - introduction / elimination rules for $-$ (denial)

The signed-formula approach

Seminal Paper: Rumfitt (2000): 'Yes' and 'No'

$$\begin{array}{c}
 \frac{-A}{-(A \wedge B)} \quad -\wedge I_1 \qquad \frac{-B}{-(A \wedge B)} \quad -\wedge I_2 \qquad \frac{-(A \wedge B) \quad \begin{array}{c} [-A] \\ \vdots \\ C \end{array} \quad \begin{array}{c} [-B] \\ \vdots \\ C \end{array}}{C} \quad -\wedge E \\
 \\
 \frac{-A \quad -B}{-(A \vee B)} \quad -\vee I \qquad \frac{-(A \vee B)}{-A} \quad -\vee E_1 \qquad \frac{-(A \vee B)}{-B} \quad -\vee E_2 \\
 \\
 \frac{+A \quad -B}{-(A \rightarrow B)} \quad -\rightarrow I \qquad \frac{-(A \rightarrow B)}{+A} \quad -\rightarrow E_1 \qquad \frac{-(A \rightarrow B)}{-B} \quad -\rightarrow E_2 \\
 \\
 \frac{-A}{+(\neg A)} \quad +\neg I \qquad \frac{+(\neg A)}{-A} \quad +\neg E \qquad \frac{+A}{-(\neg A)} \quad -\neg I \qquad \frac{-(\neg A)}{+A} \quad -\neg E
 \end{array}$$

The signed-formula approach: Perfect harmony?

- Rumfitt (2000, p. 806): the rules “exhibit a kind of harmony”
- del Valle-Inclan & Schlöder (2023): “by all known standards of harmony, these rules are harmonious — trivially, what one gets from an elimination is what is required for an introduction”
- Bonus claim: on this picture it is the *intuitionistic* rules that look anomalous — classical logic is primary

Beyond the Operational Rules: Coordination Principles

Operational rules alone are not enough to fix classical logic. We also need structural principles governing how assertion and denial interact:

C_1

If $\Gamma \vdash \alpha$ and $\Gamma \vdash \alpha^*$, then $\Gamma \vdash \perp$.

C_2 (Smilean reductio)

If $\Gamma, \beta \vdash \perp$, then $\Gamma \vdash \beta^*$.

α^* : converse sign of α . Lower-case Greek: signed formulas; capital Greek: sets thereof.

- $C_1 \approx$ law of non-contradiction (exclusivity of assertion and denial)
- $C_2 \approx$ classical reductio (exhaustivity of assertion and denial)
- Rumfitt (2002): needed to guarantee a *coherent* bilateral sense for the connectives.

Some Objections

Gabbay (2017): Harmony $\neq$ Consistency

- Constructs a connective that
 - is **harmonious** by Rumfitt's own lights, yet
 - **trivializes** the system (a ~~tonk~~-like connective)
- Conclusion: in a bilateral framework, harmony of a connective's rules does *not* guarantee consistency
- **Francez (2018)** replies:
 - Gabbay used the wrong notion of harmony
 - A bilateral system needs harmony
 - ⇒ Along the introduction/elimination axis, *and*
 - ⇒ Along the assertion/denial axis

Some Objections

Ferreira (2008): Are the coordination principles properly justified?

- Shows: C_2 does not preserve from atomic to complex instances
- ⇒ Stipulating C_2 for atomic sentences is *not* enough to derive it for complex ones
- C_2 must be separately postulated for complex sentences too
- Problem: then the sense of a connective (e.g. negation) is **not fully determined** by its operational rules alone – bad news for a bilateralist who wants meaning fixed by inference rules

Some Objections

Rumfitt's Reply (2008)

- At least C_1 is preserved by the structural/operational rules alone – this is important
- For C_2 : Yes, just postulate it for arbitrary (complex) sentences

“If a sentence has a determinate bilateral sense, then the grounds for asserting it must not intersect the grounds for denying it. But it is no requirement of determinacy of bilateral sense that any grounds that contradict the denial are ipso facto grounds for assertion, as (C_2) requires.”

(Rumfitt 2008, p. 1061)

Upshot: C_1 has priority C_2 for Rumfitt

Some Objections

del Valle-Inclan & Schlöder (2023): Pushing further – no harmony for Rumfitt's system

- Connecting harmony directly to Ferreira's coordination concern
- Present a *tonk*-like connective whose rules:
 - Are harmonious (assertion/denial & I/E pairs match up), **and**
 - Do preserve C_1 (Rumfitt's requirement!),
 yet yielding a **trivial** system
- Diagnosis: harmony of the operational rules and preservation of the co-ordination principles must be shown *together* — not harmony of the rules in isolation

A genuinely **bilateral notion of harmony**:

Unilateral harmony for assertion rules and for denial rules, *plus* preservation of all co-ordination principles.

Some Objections

del Valle-Inclan & Schlöder (2023): Pushing further – no harmony for Rumfitt's system

To remedy Rumfitt's system, they finally propose replacing the rules for rejection of conjunction and assertion of disjunction by the following ones:

$$\begin{array}{c}
 [+A] \\
 \vdots \\
 \frac{-B}{-(A \wedge B)} \quad -\wedge I \quad \frac{-(A \wedge B) \quad +A}{-B} \quad -\wedge E
 \end{array}$$

$$\begin{array}{c}
 [-A] \\
 \vdots \\
 \frac{+B}{+(A \vee B)} \quad +\vee I \quad \frac{+(A \vee B) \quad -A}{+B} \quad +\vee E
 \end{array}$$

The coherence-based approach: A bilateral reading of the sequent calculus

Back to the debate about intuitionistic vs. classical logic

Why is PTS leaning towards intuitionistic logic?

- **Preference for natural deduction systems**, which for IL satisfy certain PTS-desirable properties that CL systems do not
- Another way to answer this: Why not consider sequent calculus systems instead?
- Remember: The classical sequent calculus has the same good features as the single-conclusion calculus
- Problem: Is reasoning with multiple conclusions natural?

The coherence-based approach

Dummett's criticism of multi-succedent SC

Starting Point: Dummett's (and others') critique of CL with respect to multiple conclusions in sequent calculus:

Sequents with two or more sentences in the succedent [...] have no straightforwardly intelligible meaning, explicable without recourse to any logical constant. (Dummett 1991, p. 187)

- Asserting A and asserting B is tantamount to asserting " A and B "
- ⇒ Although formulas in antecedent are in a sense conjunctively connected, we can understand the meaning of a sequent with more than one formula in antecedent without having to have a prior understanding of "and"
- With multiple formulas in succedent it's different: they are disjunctively connected and we cannot understand these sequents without a prior understanding of "or"

The coherence-based approach

Dummett's criticism of multi-succedent SC

How can we explain a sequent of the form $A \vdash B, C$?

- If you have asserted A , you may assert either B or C ?
- ⇒ No, it's not a random choice
- If you have asserted A , then either you may assert B or you may assert C ?
- ⇒ Unhelpful until you know which one is safe to assert
- If you have asserted A , you may assert " B or C "?
- ⇒ Yes, but for that you're presuming an understanding of the constant "or"

The coherence-based approach

Seminal Paper: Restall (2005): Multiple Conclusions

Restall against this:

- Based on wrong conception of validity as warrant-to-assert preserving
- Proposes a different understanding of sequent calculus:
 - Validity of $\Gamma \vdash \Delta$ understood as an *incoherent* position of asserting each of the members of Γ while denying each of the members of Δ
 - Other terminology: position clashes, position is ‘out of bounds’
 - (N.b.: Only later this position was tagged as ‘bilateralism’)
- A conclusion follows from premises iff asserting all premises *and* denying the conclusion is incoherent
- Underlying thought: An utterance may be acceptable alone but clash with other commitments

The coherence-based approach

Seminal Paper: Restall (2005): Multiple Conclusions

Why does this solve Dummett's problem?

Dummett's objection against classical SC

We read the left side of the sequent conjunctive, the right side disjunctive – only possible with a preceding understanding of disjunction.

Restall's solution

With a bilateral interpretation, we read *both* sides conjunctive:

$\Gamma \vdash \Delta$ valid iff asserting *all* formulas in Γ while denying *all* formulas in Δ is incoherent.

Reflexivity, Contraction & Weakening

- Reflexivity: asserting and denying the same formula is incoherent
- Contraction: if asserting (or denying) a sentence twice is incoherent, so is doing it once
- ⇒ Plausible: number of repetitions irrelevant; content causes the clash
- Weakening: If a set of assertions/denials is already incoherent, adding more cannot restore coherence

Cut: A difficult case?

- From $\Gamma \vdash A$ and $\Gamma, A \vdash B$, infer $\Gamma \vdash B$
 - Out-of-bounds reading: if asserting Γ and denying A , while asserting A and denying B is incoherent, then asserting Γ while denying B alone is incoherent
 - We are incoherent no matter what A is
 - Cut imposes a requirement to be able to either assert or deny (any!) A
- ⇒ Ripley (2013, 2015) Unjustified!
- Sometimes A itself is the source of incoherence
 - Paradoxical sentences (e.g. the Liar) cannot be consistently asserted *or* denied
 - Therefore cut need not hold universally (tonk is therefore okay too!)
- ⇒ Offers a motivation for a substructural approach to paradoxes

A proof-theoretic analysis of paradoxes

Paradoxes – proof-theoretic definition (Tennant 1982)

- Non-normalizable proof of $\perp$
- (Or, in case of Curry's paradox, of an arbitrary atomic formula)
- ⇒ Normalization procedure does not terminate but enters an oscillating loop or infinite regress (Yablo's paradox)
- In SC: no cut-free version of the proof available

Let λ be the name of a sentence saying about itself that it is false and extend our language with a transparent truth predicate

$\lambda := \neg T\lambda \ (T\lambda \rightarrow \perp)$

Rules for the λ -constant:

$$\frac{\Gamma \vdash T\lambda \rightarrow \perp}{\Gamma \vdash \lambda} \lambda R \qquad \frac{\Gamma, T\lambda \rightarrow \perp \vdash A}{\Gamma, \lambda \vdash A} \lambda L$$

The Liar in sequent calculus

$$\begin{array}{c}
 \frac{\overline{\lambda \vdash \lambda} \text{ } Rf}{\frac{\frac{\overline{\lambda \vdash \lambda} \text{ } Rf}{T\lambda \vdash T\lambda} \text{ } TL/R}{T\lambda, \neg T\lambda \vdash \perp} \neg L} \quad \frac{\overline{\lambda \vdash \lambda} \text{ } Rf}{\frac{\frac{\overline{\lambda \vdash \lambda} \text{ } Rf}{\lambda \vdash T\lambda} \text{ } TR}{\lambda, \neg T\lambda \vdash \perp} \neg L} \\
 \frac{\frac{\frac{\frac{\frac{\frac{\overline{\lambda \vdash \lambda} \text{ } Rf}{T\lambda \vdash T\lambda} \text{ } TL/R}{T\lambda, \neg T\lambda \vdash \perp} \neg L}{T\lambda, \lambda \vdash \perp} \lambda L}{T\lambda, T\lambda \vdash \perp} \text{ } TL}{\frac{T\lambda \vdash \perp}{\vdash \lambda} \text{ } \lambda R} \quad \frac{\frac{\frac{\frac{\overline{\lambda \vdash \lambda} \text{ } Rf}{\lambda \vdash T\lambda} \text{ } TR}{\lambda, \neg T\lambda \vdash \perp} \neg L}{\lambda, \lambda \vdash \perp} \lambda L}{\frac{\lambda \vdash \perp}{\vdash \perp} \text{ } cut} \\
 \vdash \perp
 \end{array}$$

- In ND exclusively the meaning-giving rules of the connectives are used
- In SC additionally, structural rules are used
- Possibility to resolve paradoxes in SC:
 - Use a substructural approach
 - Reject one of the structural rules used
 - Usually focus on contraction or cut

Summary: Bilateralism in a nutshell

- Meaning and logical consequence should not merely rely on a positive concept (assertion, truth, provability) but equally on a complementary concept (denial, falsity, refutability)
- Positive and negative categories to be considered on a par, both as primitive, i.e., not reducible to each other
- In PTS: Not only assertion conditions should be considered but also denial conditions

Summary: Two strands of bilateralism

Ripley (2017, 2020) distinguishes two kinds of bilateralism:

- Warrant-based bilateralism: relevant conditions under which propositions can be warrantably asserted or denied
- Motivating a proof system of natural deduction with signed formulas for assertion and denial
- Coherence-based bilateralism: relevant conditions under which *collections* of propositions can be coherently asserted and/or denied together
- Motivating a specific kind of reading of sequent calculus
- **Both motivate a PTS approach for classical logic**

Thanks for your attention!

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Tomorrow

Non-classical frameworks

Any Questions?